

TN+NTN Deep RL Pipeline — Methodology Bug Report

Review of Hybrid_Network_TN_NTN_Final.csv, TN_NTN_Deep_RL_MODEL.ipynb, feature_TN_NTN.pdf

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Bottom line

The three new files describe a **DRL model that does not exist yet**. What actually exists is a data-cleaning/EDA notebook and a merged CSV, with several bugs that will bite as soon as someone tries to train on this data.

Bugs found

- 1. The “DRL model” notebook contains no RL model.** TN_NTN_Deep_RL_MODEL.ipynb (100 cells) is pure EDA + merging: null/dup checks, log transforms, correlation heatmaps, concatenation of the 5 source CSVs, and plots. There is no state/action/reward definition, no algorithm (DQN/PPO/etc.), no model class, no training loop.
- 2. No feature extraction or selection ever happened.** Every raw column from the 5 source CSVs is carried through untouched into the final CSV — nothing is engineered, ranked, or dropped. feature_TN_NTN.pdf is a *design table* describing intent; no code implements it.
- 3. Action-label leakage risk.** The PDF labels network_type as the DRL “Action Label,” but it also sits in the CSV as a plain feature column, identical in form to every other state feature. If the agent’s state includes network_type, the label it’s supposed to predict is already inside its own input — a degenerate policy is guaranteed unless this column is explicitly excluded from the state vector.
- 4. Categorical features are raw, unencoded strings.** Available_Networks (comma-joined string), Area, and network_type are stored as plain text. No one-hot/multi-hot encoding exists anywhere in the notebook — none of these columns are model-ready as-is.
- 5. Generated data contradicts its own design spec.** The PDF marks Rain features (“not applicable”) for 5G. The final CSV disagrees: NR_5G rows have nonzero Rain_Rate_mmhr (mean 0.40, max 16 mm/hr) and Rain_Fade_dB (max 0.02 dB). Physically negligible, but it means the shipped data doesn’t match the document describing it.

What did work

Zero-filling of not-applicable features is done correctly for the other cases: altitude_m / speed_ms are 0 for every non-UAV row, and Rain_Rate_mmhr / Rain_Fade_dB are correctly 0 for all WiFi rows — matching the PDF spec exactly, just not for the 5G case above.

Before any model training starts

1. Decide `network_type` state-vs-action split and enforce it in code.
2. One-hot/multi-hot encode `Area`, `network_type`, `Available_Networks`.
3. Define the actual state vector, action space, and reward function — none exist yet.
4. Reconcile the 5G rain-feature mismatch against the PDF spec (or update the spec).